



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 65: Solar Playground with Security System Bi-Weekly Update 4

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Kitchel, Victor Igbokwe
Sponsor: Wonhyeok Jang
TA: Zian Wang**

Project Summary

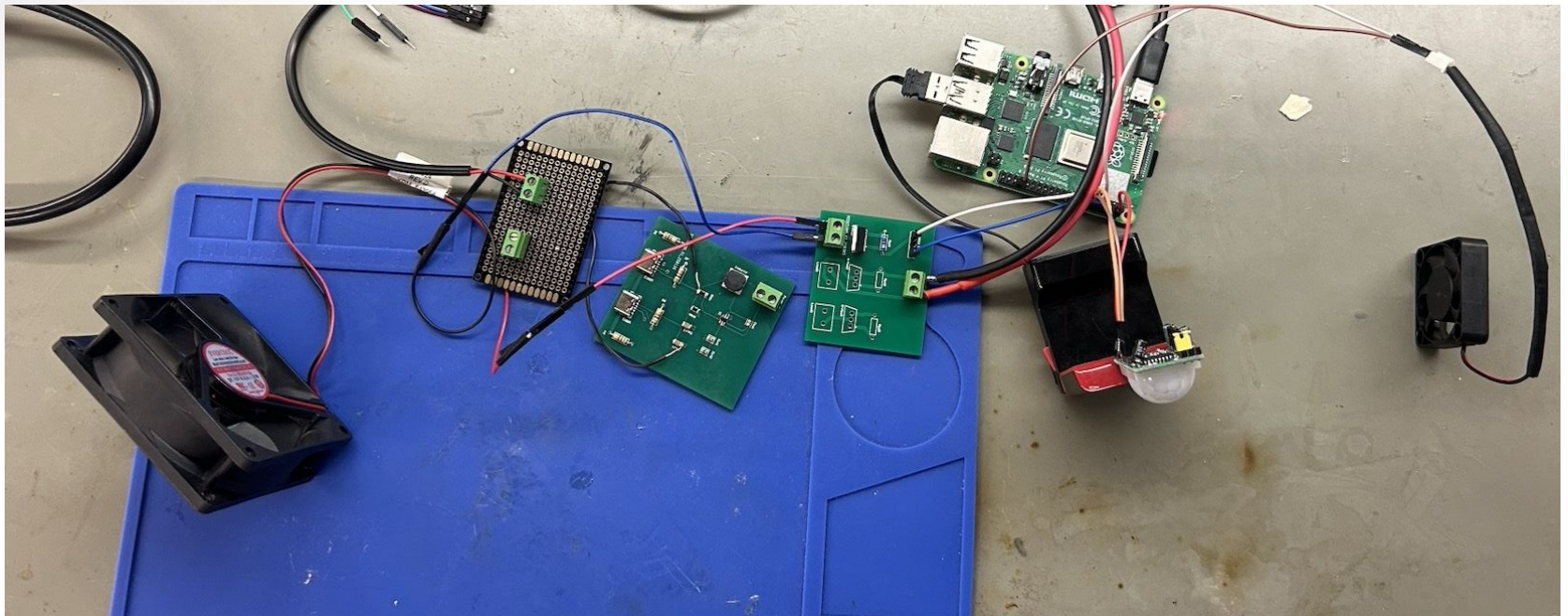
The SolarEye Security System is a solar-powered, autonomous surveillance solution for playgrounds and parks.

It enhances security with motion-activated cameras, real-time mobile alerts, LED deterrent lighting, and visitor charging stations.

This eco-friendly system ensures continuous, cost-effective protection for private parks and properties.



Integrated System Diagram (Hardware)



Integrated System Diagram (Software)

Create an Account

Minimum 6 Characters, 1 Uppercase Character, 1 Special Character

Email Address

Password

Confirm Password

CREATE NEW ACCOUNT

BACK TO LOGIN

Figure 1: User Creates Account

Hello,

Follow this link to verify your email address.

https://solareye-security-system.firebaseio.com/_/auth/action?mode=verify

If you didn't ask to verify this address, you can ignore this email.

Thanks,

Your solareye-security-system team

Your email has been verified

You can now sign in with your new account

SolarEye

zayanprasla@tamu.edu

LOGIN

CREATE NEW ACCOUNT

Figure 2: User's Email is verified and then able to login

SolarEye

SOLAREYE SECURITY SYSTEM

Live Stream

FAQ

Settings

Account Settings

SYSTEM

Figure 4: Extra options available on sidebar

5:58

SolarEye

Event History

2025-02-25

Event Type: Motion Detected
4:57 PM

Event Type: Motion Detected
4:49 PM

Figure 5: Event History Page



Figure 3: Home Page

System Controls

Figure 6: System Controls for LED, CAMERA, and SOS.

Camera Control

ON OFF

Light Control

ON OFF

Emergency Call

Are you sure you want to call 911?

NO YES



Project Timeline

New PCBs Ordered/ Received (completed 02/03)	Integration of MPU and Application subsystems (to completed 02/25)	Integration of Power Delivery system with MPU (to complete by 02/25)	Integration of Power Generation System with Power Delivery System (to complete by 03/26)	System Housing (to complete by 03/28)	Final Integration (to complete by 04/10)	Validation (to complete by 04/20)	Demo and Report (to complete by 04/25)
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Jackson Kitchel

Accomplishments since last update 30 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ PCB was redesigned to fix capacitor problem and consolidate ❖ New PCB was soldered and tested ❖ Integration with solar panel and battery complete and tested	❖ Integrate sensor outputs with MCU subsystem ❖ Start Housing ❖ Finish testing integration with power delivery subsystem

Jackson Kitchel

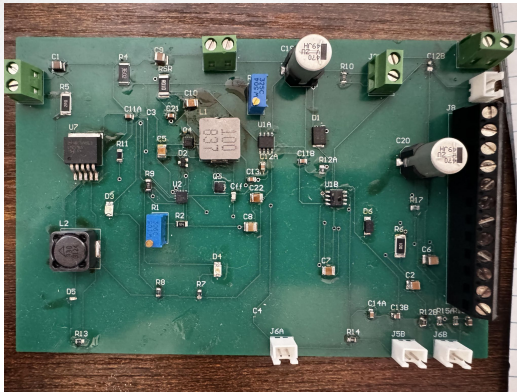


Figure 7: Old PCB Design

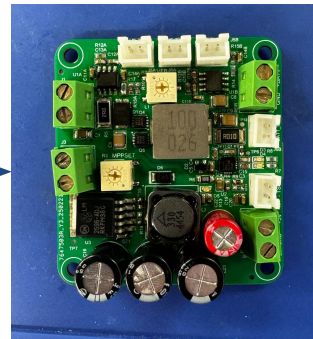


Figure 8: New PCB Design

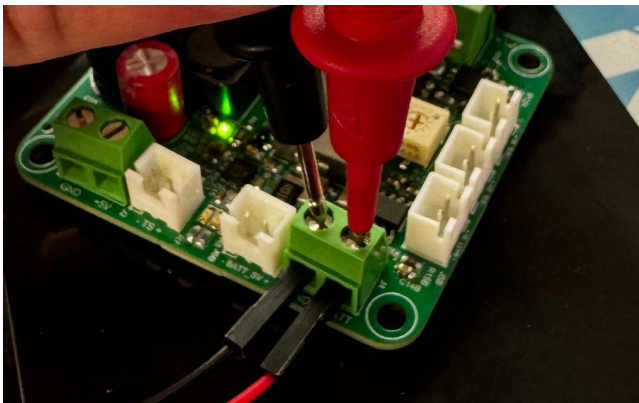


Figure 9: PCB Output Test Point



Figure 10: Charging PCB Output

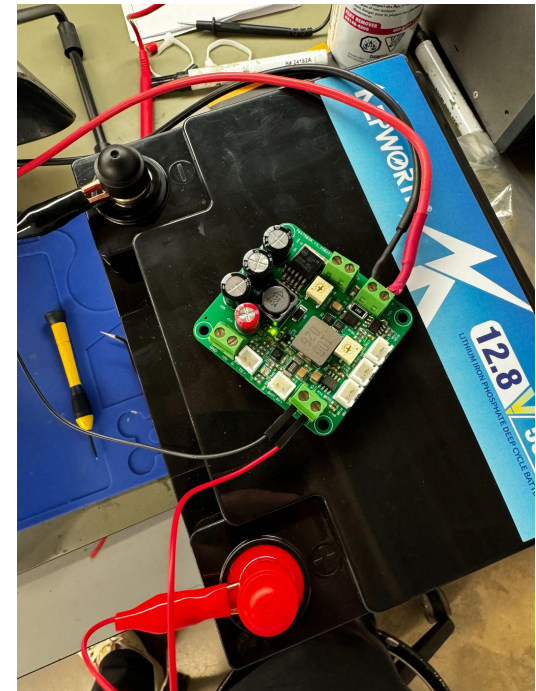


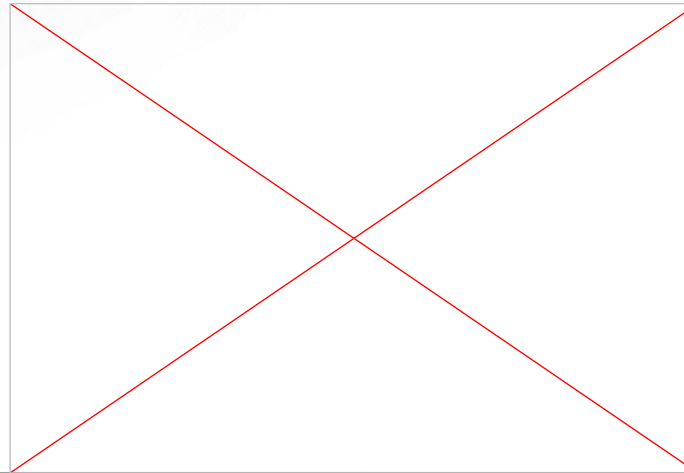
Figure 11: Battery Charging from Solar Panel (Charge LED On)



Victor Igbokwe

Accomplishments since last update 12 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Integration of Power Distribution and MPU system completed ❖ Validated power outputs for different modes of operation with MPU	❖ Integration with Power Generation system. ❖ Design/Build Housing for Circuit boards(During capstone blitz).

Victor Igbokwe



Media 1: Video showing the Buck converter supplying power to MPU while livestreaming

Mode of Operation	Power Consumption(A)
Idle	1.88 A
Event Detection	3.65 A
Live Streaming	2.15A

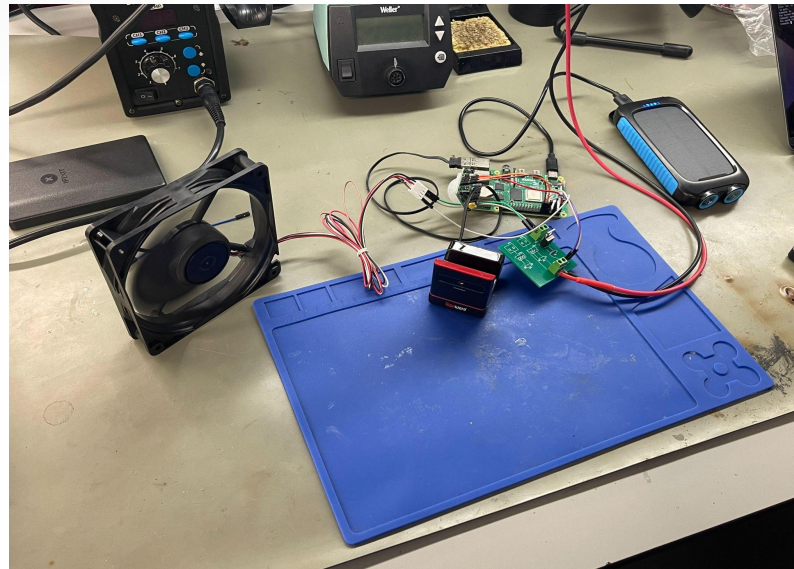
Figure 12: MPU Operating Modes and Corresponding Power Consumption Table



Alex Allahar

Accomplishments since last update 8 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Integrated Relay Controller PCB with MPU. ❖ Validate Event History on App Database ❖ Verified MPU / APP controls LED and cooling fans.	❖ Validate Event History on Mobile App ❖ Start housing during Blitz.

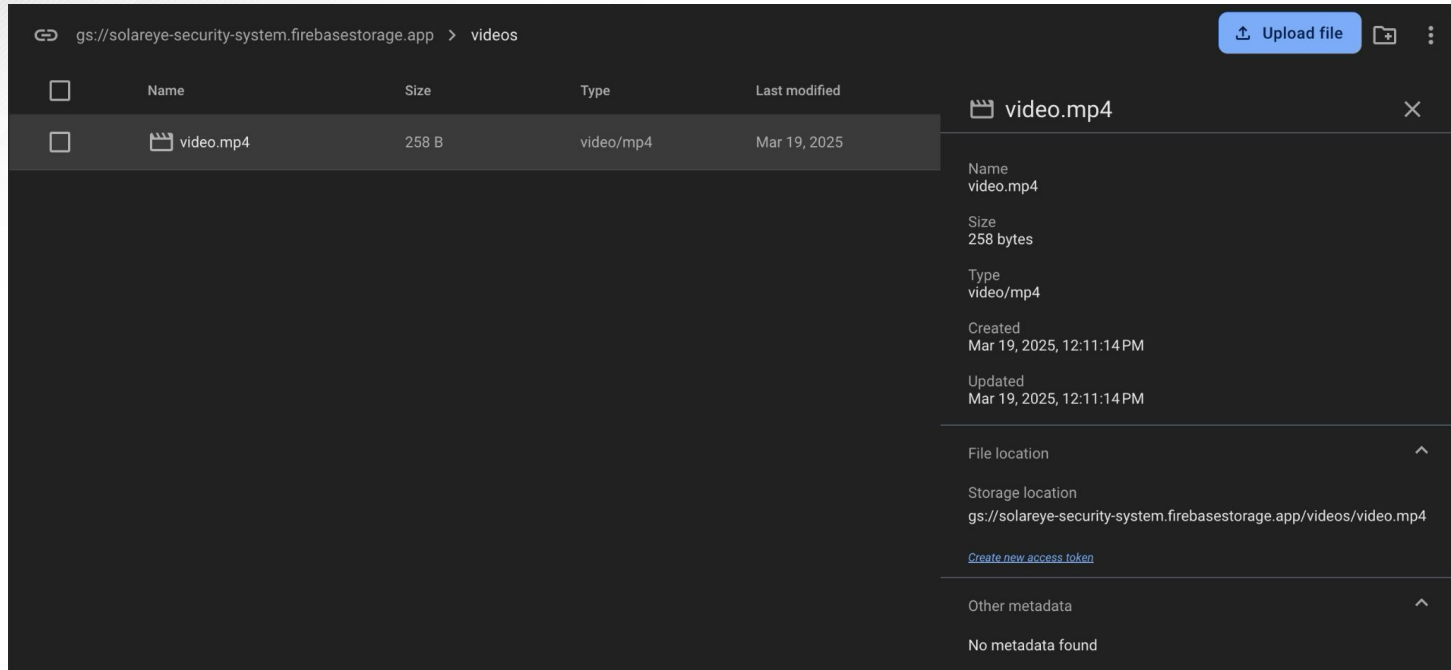
Alex Allahar



Component	Power Consumption(A)
Small Fan (Fixed)	0.6 A
Medium Fan (Fixed)	1.16 A
Large Fan (Variable)	7.017 A (100 %)

Figure 13: Fan Component Testing Setup and Table

Alex Allahar



```
pi@raspberrypi:~/solareye $ python video_firestore.py
File uploaded successfully.
pi@raspberrypi:~/solareye $
```

Figure 14: MPU Video Footage Transfer to App Firestore Database



Zayan Prasla

Accomplishments since last update 7 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Validated all button communication from App to MCU ❖ Continuous tweaks to the frontend to for smoothest app experience	❖ Validate thumbnail (picture)/video transfer from mcu to app ❖ Validate live streaming ❖ Power level communication ❖ Validate Low power mode

Zayan Prasla

+ Start collection	+ Add document	+ Start collection
camera_control	led_state >	+ Add field
⋮ led_control >		state: 1
pir_sensor_data		

Figure 15: LED Control Validation

+ Start collection	+ Add document	+ Start collection
camera_control >	⋮ camera_state >	+ Add field
led_control		state: false
pir_sensor_data		

Figure 16: Camera Control Validation

📁 pir_sensor_data	📄 7obo2W0zxmDY5xK7wuy0
+ Add document	+ Start collection
0kTB0V3J3PNJDrJeitoC	+ Add field
0rz7Ir4JVI9d4pjkyF3Z	motion: "detected"
⋮ 7obo2W0zxmDY5xK7wuy0 >	timestamp: "2025-02-25 16:49:28"
nZNjMf4xkNmLQdMZfKpS	
trTHLusozTBGLAIkRAsG	

Figure 17: Event History Database

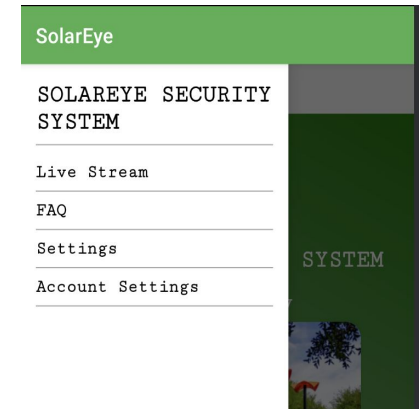


Figure 18: New Sidebar for better user accessibility

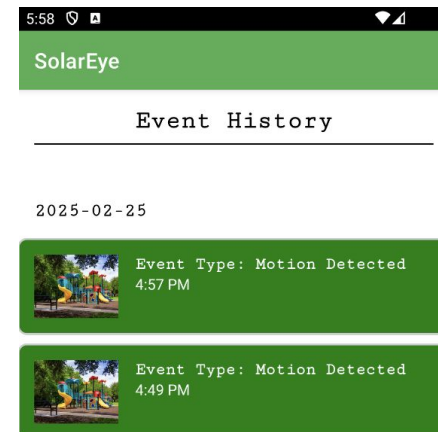


Figure 19: Integrated Event History Page

Execution Plan

[illegible]



SYSTEM Validation Plan

Validation Plan 404				
Test Name	Success Criteria	Methodology	Status	Assigned to
LED Relay Controller MPU Control	state, or automatically if motion is detected from the MPU.	between video capture and firebase acknowledgement.	Completed	Alex Allahar
Modes of Operation App Control	Be able to control all three modes of operation via the app	Using firestore database and GCP functions to send a update to the MPU changing the mode of operation.	Pending	Alex Allahar
App Event History	Both MPU and App has up to a week of footage.	Verify that the MPU and Firebase after reboot has the up to 24 of prevent footage.	Partial Completion	Alex Allahar
Video Streaming	The MPU live-stream can be access via App.	Use UDP to Stream the MPU via HTTP Flask Server.	Partial Completion	Alex Allahar
MPU to App Automatic Delay	Delay is below 1 s. To make it almost automatic.	Using PIR_sensor_data_firestore.py perform 10 testcases and average the delay.	Completed	Alex Allahar
Charging Statistics to App	Communicate in less that 1s the Charging Statistics from the Power Generation PCB.	Use GPIO pins to gather the current, and voltage data of the battery to determine the system uptime and communicate this information with Firestore.	Pending	Alex Allahar
Validate Buck integration with MPU	All modes of operation on MPU function while integrated wit buck converter PCB	Connect PCB and test all modes of operation	Completed	Victor Igbokwe
Buck LineTest	Output voltage remains constant with different input voltages	Change input voltage and record output voltages	Completed	Victor Igbokwe
Buck Load Test	Output voltage remains constant as the output current increases.	Use Electronic load device	Completed	Victor Igbokwe
Relay control module Integration	Current flow can be controlled by MPU	Connect GPIO pins on PCB to MPU	Completed	Victor Igbokwe
Buck Converter integration with Solar Generation subsystem	Buck converter powered by solar generation system	Connect buck PCB to battery	Pending	Victor Igbokwe
Network Connectivity	Able to receive and send signals to and from the app/system.	MPU data is able to be sent and seen on the app.	Completed	Zayan Prasla
Password Policy	User has to put in a password that follows the password policy	Check if password can be created by not following the policy: 6 minimum characters, 1 uppercase and 1 special character	Completed	Zayan Prasla
Store and Fetch Data	Application is able to store and fetch login data.	Test logging into the app with an already created account	Completed	Zayan Prasla
Email Verification	After creating an account, user receives a verification email.	Check for a verification email being sent after creating an account	Completed	Zayan Prasla
Main Page UI	Successfully display Event History, Charging statistics, and system controls	Check each function on main page and make sure there is no input lag	Completed	Zayan Prasla
Live Streaming	Live stream footage directly from the system camera to the app	Check if live stream is not too behind from actual live camera time	Pending	Zayan Prasla
Low Power Mode	Turn on lower power mode with a switch on the app which communicates with the mpu	Check if mpu switches to low power mode once applied from app	Pending	Zayan Prasla
Video/Picture Transfer	Successfully transfer pictures and videos with the correct date/time	Check the event history page and make sure picture and video align with date and time	Pending	Zayan Prasla
Solar Generation Check	A charge rate of 0.45kW.	Test solar charge rate with a multimeter.	Completed	Jackson Kitchel
Operation Time	8 hours of battery life	Monitor the battery after it finishes fulling charging.	Pending	Jackson Kitchel
High Voltage Cut-Off	Charging circuit cuts off when voltage level reaches 15V	Monitor the charging voltage after battery is fully charged	Completed	Jackson Kitchel
Low Voltage Cut-Off	Battery starts charging when low-voltage of 12.4V	Monitor the charging voltage after charge level is low	Completed	Jackson Kitchel
PCB Functionality	PCB Soldered and functioning as required by previous checks	Validate every requirement after PCB is complete	Completed	Jackson Kitchel
PCB Design	PCB Design is plausible.	Simulation Results using software	Completed	Jackson Kitchel
System Housing	The System is able to function and hold all the needed subsystems.	Test All Modes of Operation with the completed housing for the system installed.	Pending	ALL
Complete system Test	The complete system with all working subsystems is able to function as expected	Test All Modes of Operation with the complete system	Pending	ALL



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Thank You!